Supplementary Notes S1–S5

S1 Exploratory Tier-1 land sector • S2 Parameter and factor provenance • S3 Uncertainty inputs • S4 Implied buffalo factor • S5 QA/QC and the audited nitrogen chain

Companion to "A Reproducible National Agricultural Greenhouse-Gas and Air-Pollutant Inventory with Mass-Conserving Geospatial Allocation for Pakistan"

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Release provenance. This document describes PakEmis under the period, configuration and source boundaries specified in its methods. The identifiers below link it to the accompanying products and source register.

Release version	`1.0.0rc69`
Commit	`d8d874622f9a`
Reporting period	July 2015–June 2025 (ten fiscal years) (2015-07-01 to 2025-06-30)
Reference	S_TAN (household survey; TAN nitrogen)
FY2023–24 total	444.61 Mt CO_2e (AR5) under S_TAN, the reference; 449.85 Mt under L_TAN
Release manifest	`ac0ffe93ad9fcd88` (SHA-256 of `outputs/release_manifest.json`, first 16)
Source manifest	`c1be6d314cfce1ea` (SHA-256 of `data/manifests/source_manifest.csv`, first 16)
Data-description article	`paper_national/` (ESSD, in preparation), same release

S1.1 Overview and scope

This Supplementary Note documents an exploratory Tier-1 land-use, land-use change and forestry (LULUCF) extension of the agricultural inventory. It is *excluded from every agricultural total, benchmark comparison, key-category analysis, and uncertainty estimate* in the main paper; we report it here for completeness because Pakistan's land sector is frequently cited in national climate discussion, but its data basis is weaker than the agricultural inventory and its result should be read as indicative only.

S1.2 Method

The land sector is computed separately from agriculture and is dominated by removals. Forest carbon is estimated as a net afforestation sink consistent with the FAO Forest Resources Assessment forest-land series (FAO, 2020); Indus-delta mangrove blue carbon uses Global Mangrove Watch (GMW) area (Bunting et al., 2022) and 2013 Wetlands Supplement carbon densities (IPCC, 2014); cropland and grassland soil-carbon change are not estimated at Tier 1 (notation key NE, contributing nothing to the net rather than a zero). The estimate is single-year.

S1.3 Result and its uncertainty

A first-order Tier 1 calculation gives a small net sink of order −4 Mt CO_2e (forest plus Indus-delta mangrove blue carbon). Both its magnitude and its *sign* are uncertain: the humid-tropical default carbon densities likely overstate storage in the arid delta, and although Pakistan is often

cited for mangrove afforestation, GMW v3 indicates a *recent decline* in delta mangrove extent, which would shrink or reverse the mangrove sink relative to the afforestation baseline. The land factors used are summarised in Table 1; a properly developed land sector — with forest-area time series, stock-change factors, separated biomass and soil pools, and its own uncertainty analysis — is future work and is not attempted here.

Land category	**Basis**	**Notation**
Forest land	FAO FRA forest-land series	E
Cropland / grassland (SOC)	zero at Tier 1	NE
Wetlands (mangrove)	restored area × 6.5 tCO_2/ha/yr (flux)	E

Table 1: Land-sector factors and notation keys (E estimated, NE not estimated). The mangrove row is an annual FLUX—restored area at a sequestration rate—never area × carbon density, which is a standing STOCK (the 471 $t\,ha^{-1}$ humid-tropical soil-carbon default puts the stock near 240 Mt CO_2e, reported in the engine's separate stock column and not as a removal). The stock–flux distinction is the reason the note's sink is of order −0.5 Mt/yr for mangroves, not hundreds.

S1.4 Reproducibility

The land sector is an optional module (`src/pkinv/lulucf/`) and is *not* part of the default agricultural reproduction run; it is skipped under `run_all.py -quick` and its outputs (`lulucf_totals.csv`) feed only this note, never the agricultural totals or analyses of the main paper.

S2.1 Purpose

This note gives manuscript-level provenance for every country-specific parameter and emission/scaling factor the main paper relies on: value, status, source, and (where relevant) derivation. Every value is version-controlled in the corresponding `data/*.csv` file with the same `source` string. Values marked *assumption* are structured judgments made in the absence of a measured Pakistan-specific value and are examined by the sensitivity analysis where they matter (main paper, Table 5).

S2.2 Livestock characterisation and activity

Table 2 lists the species head-counts (activity) and the Tier-2 characterisation parameters (typical animal mass TAM and nitrogen-excretion rate Nex) that drive enteric and manure emissions. Per-head N excretion is $Nex = Nrate \cdot TAM \cdot 365/1000$ (kg N/head/yr); the total N excreted (≈7.9 Mt N/yr) is the activity basis for manure-management N_2O.

Species	Head (M)	TAM (kg)	Nrate	Source
Dairy cattle	16.48	400	0.47	PBS census 2024; LRRD 2006 (TAM); IPCC T10.19 (Nrate)
Other cattle	39.38	225	0.34	PBS census 2024; Pak. zebu avg.; IPCC T10.19
Buffalo	47.74	380	0.32	PBS census 2024; Nili-Ravi herd avg.; IPCC T10.19
Sheep	44.58	35	1.17	PBS census 2024; IPCC T10.19
Goats	95.83	32	1.37	PBS census 2024; IPCC T10.19
Camels	1.51	480	0.46	PBS census 2024; Khan 2025; IPCC T10.19
Poultry	2065	1.8	0.82	Economic Survey 2023–24; IPCC T10.19

Table 2: Livestock activity (head-counts) and Tier-2 characterisation. Nrate is kg N/1000 kg/day. Dairy share 0.295 (FAOSTAT QCL 2024) splits cattle into dairy/other.

S2.3 Manure management-system shares and dung-cake

Management-system (MMS) shares (`data/mms_shares.csv`): cattle — pasture 0.30, daily-spread 0.17, solid-storage 0.15, drylot 0.10, *burned as fuel 0.28*; buffalo — pasture 0.20, daily-spread 0.20, solid-storage 0.20, drylot 0.10, *burned as fuel 0.30*; sheep/goats/camels — pasture 0.90, solid-storage 0.10; poultry — litter 1.00. Sources: South-Asia livestock MMS literature (AAQR 2024) and Tier-2 estimates; the dung-cake (burned-fuel) fractions are from Pakistan dung-cake fuel studies (MDPI *Sustainability* 2021), and that nitrogen is transferred to Energy (IE), not counted in agriculture.

S2.4 Soil N_2O factors and nitrogen fractions

From `data/emission_factors.csv` (2019 Refinement V4 Ch11 unless noted): EF_1(dry) 0.005, EF_{1FR}(rice) 0.004, EF_{3PRP} 0.002 (cattle)/0.003 (sheep-goat, 2006 T11.1), EF_4 0.005, EF_5 0.011; $f_{NH_3,SF}$ 0.1608 (product-weighted from the NFDC mix, superseding the IPCC default; FracGASF proper, the NH_3+NO_x volatilised pool of Eq. 11.9, is 0.1678), FracGASM 0.21, FracLEACH 0.24 (T11.3). Nitrogen fractions (`data/parameters.csv`): urea share of applied N 0.80 (NFDC), rice share of synthetic N 0.16 (area-weighted), irrigated fraction 0.80 (FAO AQUASTAT) applied to FracLEACH on canal-irrigated land. The aggregate EF_1 0.010 (2006) and wet EF_1 0.016 are retained for sensitivity only; the empirical anchor is Cui et al. (2021).

S2.5 Residue, burning, and rice factors

Residue ratios and returned fractions (`data/residue_params.csv`; IPCC ratios + Pakistan residue management, IGC): above-ground residue ratio wheat 1.10 / rice 1.00 / maize 1.00 / sugarcane 0.25 / cotton 2.00; fraction returned 0.20–0.30 (dry matter is folded into these ratios). Burning (`data/burning_params.csv`, `emission_factors.csv`): burned fractions rice 0.30, wheat 0.05, cotton 0.10, maize 0.05, sugarcane 0.10 (IGC Pakistan); combustion factor C_f 0.85 (IPCC 2006 T2.6); G_{ef} 2.7 g CH_4/kg d.m., 0.07 g N_2O/kg d.m. (T2.5). Rice (`emission_factors.csv`, IPCC 2006 V4 Ch5): baseline EF_c 1.30 kg/ha/day (T5.11), SF_w 1.0 continuous default (T5.12; intermittent 0.52, rainfed 0.28 for sensitivity), SF_p 1.0 (T5.13), SF_o 1.0 (no organic amendment, Eq. 5.3), cultivation period 125 days (PARC calendar, area-weighted).

Status of the provenance strings, classified. The IPCC factors, the empirical soil-nitrogen anchor (Cui et al., 2021), the EMEP/EEA manure factors (EMEP/EEA, 2019; Sommer et al., 2019) and the biomass-burning factors (Akagi et al., 2011) are formally cited in the main paper. The country-specific parameters are not all of one kind, and a previous edition of this note described them in prose that named five families and silently omitted a sixth. Every affected row now carries a `provenance_class` column, so the distinction is machine-readable and countable rather than a paragraph a reader has to trust. Across the five parameter files (59 rows):

- **10 cited** — resolve to a retrievable reference: the IPCC Table 10.19 nitrogen-excretion rates, the IPCC poultry mass default, FAOSTAT stocks and the FAO AQUASTAT irrigated fraction.
- **3 derived** — computed from another parameter or from a definition (urea nitrogen content is stoichiometric; the leaching-applicable fraction is the irrigated fraction), so no external source applies.
- **25 working pointers** — the source string names a body of literature but not a specific author, title and volume: `"South-Asia livestock MMS literature (AAQR 2024)"`, `"Pakistan dung-cake fuel studies (MDPI Sustainability 2021)"`, `"Pak Dairy Info; LRRD 2006"`, `"Khan 2025"`, `"PARC rice calendar"`, `"IGC Pakistan"`, `"NFDC"`, `"Climate research"`. They identify where a value came from without letting a reader retrieve it.
- **21 declared judgements** — no external source at all: `"Tier 2 estimate"`, `"Predominantly grazing"`, `"Range/extensive"`, the population-average animal masses, and the two liming rows marked `ASSUMPTION`. These are assumptions of this inventory, and are now recorded as such rather than sitting in the same column as a citation.

Two consequences are worth stating plainly. The manure-management shares are entirely in the last two classes — fourteen declared judgements and three working pointers, none cited — and two of those rows, the cattle and buffalo `burned_fuel` fractions, carry the cross-sector dung transfer to residential fuel. Configuration L therefore rests on two rows that name no document. And resolution was attempted for this release and *not* completed: identifying which specific paper a value came from requires knowing what the original reader had in front of them, and choosing a plausible-looking match would fabricate provenance rather than establish it. The gap register carries this as `agriculture-parameter-provenance`; the honest position is that these are declared inputs with a stated class, not references, and the count above is the size of the debt.

S3.1 Conditional parameter calculation

The current annual conditional product reruns native agricultural quantities from shared physical inputs. It does not sample unrelated category totals and then infer a material balance from their sum. Each species' sampled herd enters enteric methane, manure nitrogen and the linked ledger-L household dung quantity. Each crop production draw enters its residue destinations; fertiliser nutrient and product inputs retain their own source uncertainty and units. Synthetic fertiliser NH_3 factors retain the adopted product mix and receive one common table-wide factor multiplier in this conditional case; this is not a measurement of cross-product factor covariance. Other dependence interpretations remain an uncertainty limitation.

Nitrogen calculations retain the selected TAN or complete IPCC formulation. The same surviving material state supplies management and post-application NH_3, soil NO_x and direct/indirect N_2O. Fractions drawn within this conditional model retain their physical bounds. Management-system shares and structural factor applicability are held within a case rather than independently multiplied outside a partition. Excretion rates and unsupported management changes remain stated assumptions. A wider interval does not repair a transcription error or represent an unestimated source.

The reference assigns 0.95 of volatilized nitrogen to NH_3 under the adopted chemical attribution. A separate deterministic case assigns all volatilized nitrogen to NH_3 and reruns the linked NH_3, NO_x and N_2O pathways from the same native ledger. It tests attribution sensitivity without assigning a probability or interpreting the endpoint as an uncertainty bound. The 0.80 irrigated/leaching applicability setting is likewise a conditional assumption; approximate national area totals do not independently establish that fraction.

Input	Shared calculation	Conditional treatment
Livestock heads	Enteric, manure and ledger-L dung	Same species/input realization; census bridge held
Crop production and area	Residue flows, burning, crop nitrogen and rice	Own input uncertainty; source vintage and calendar retained
Fertiliser nutrients/products	Synthetic NH_3, soil NO_x and N_2O	Common input identity with pathway-specific factors
Nitrogen pathway factors	Management, field and indirect losses	Declared factor uncertainty and bounded fractions; material closure
Rice regime and soil climate	Methane and managed-soil factors	Conditional reference or explicit structural sensitivity
Management and destination shares	Manure, residue and dung fate	Fixed supported case or separate configuration; no invented probabilities

Table 3: Physical input sharing in the current conditional calculation. The parameter catalogue gives exact widths, distributions, means, provenance, identifiers, seed and draw count. These are separate from structural choices and from the older aggregate Approach 1 screen.

S3.2 Temporal dependence and fitness

Historical changes use paired realizations. A persistent-parameter case retains common uncertain factors across years, while a separate annual-activity-error case tests a different activity dependence. Each change is calculated within a realization before percentiles are taken. Subtracting unrelated annual percentiles would give a different quantity and is not the reported trend interval.

The conditional intervals cover admitted sources under a named configuration. Unestimated categories, alternative livestock or management reconstructions, controls, destination choices and temporal/spatial allocation uncertainty are not combined into a probability interval. The

reference uncertainty year remains FY2023–24 even though supported annual activity extends through FY2024–25. The earlier component-lognormal sector diagnostic and Approach 1 calculations retain their own settings when shown; they are not substituted for this native annual product.

1 Supplementary Note S4: the benchmark's implied buffalo factor

The South Asian ammonia inventory we benchmark against (Tomlinson et al., 2026) reports implicit emission factors by livestock type, but its table omits buffalo — the species its own text names as the reason Pakistan's grazing emissions are high, and the species that dominates Pakistani nitrogen excretion. Its released dataset is gridded emissions only (Tomlinson et al., 2025), so the underlying animal masses, excretion rates and management shares cannot be recovered from it directly. A residual back-calculation is therefore the only available route, and this note sets out what it does and does not establish.

Method. We apply the benchmark's own tabulated factors to the same FAOSTAT 2015 populations our historical series uses, subtract the result from its published Pakistan livestock total (grazing plus manure management, 495.7 kt), and attribute the residual to the animals its table does not cover.

Term	**kt NH_3**
Benchmark Pakistan livestock total (grazing + manure management)	495.7
less broiler chickens	−102.4
less dairy cattle	−77.0
less goats	−75.9
less non-dairy cattle	−60.2
less sheep	−42.4
Residual, unattributed by the published table	**137.8**
Unattributed population (buffalo + camels)	36.6 M head
Implied factor over buffalo and camels	3.76
Implied buffalo factor, upper bound	**3.87**
Their tabulated dairy-cattle factor, for scale	6.33

Table 4: Back-calculated buffalo emission factor (kg NH_3 per head per year). *This is our derivation, not a published value.* The residual also absorbs camels and any category the published table omits, so it is an upper bound on the buffalo factor alone. Generated by `src/pkinv/eval/buffalo_implied_ef.py`.

What it indicates. An implied factor of at most 3.9 kg per head sits *below* the benchmark's own dairy-cattle value of 6.33, for an animal with substantially higher nitrogen excretion. The only configuration that reproduces such a value is one in which buffalo manure is overwhelmingly deposited at pasture, where ammonia losses are small, rather than passing through housing, storage and field application. Our own species-resolved chain gives 9.83 per head and the GLEAM 3.0 South Asian management split — which assumes no buffalo grazing at all — gives 11.69. Our value therefore lies between two external parameterisations that disagree with each other by a factor of three.

What it does not establish. The calculation is a residual, so it inherits every difference between our activity data and theirs, not only the management split. It cannot separate buffalo from camels. And the two external parameterisations that bracket our value are not statistically independent — both draw on overlapping FAO activity data — so they bound the plausible range rather than corroborating each other. The conclusion this supports is that the manure-management partition is the most likely locus of the residual livestock divergence, which is why a national manure-management survey is identified in the main text's limitations as the highest-

value data collection this work points to. It is not an attribution, and no result in the main text depends on it.

2 Supplementary Note S5: quality assurance and the audited nitrogen chain

This appendix preserves the historical IPCC-chain correction used in the prior release. The numerical before/after examples below describe that diagnostic, not the current TAN reference or its conditional interval. Current nitrogen records replay each named formulation from the same adopted activity and material state; historical correction values are checked against the preserved rc67 ledger rather than used as targets for the current model.

The greenhouse-gas and air-pollutant layers of this work draw on the same excreted nitrogen. That sharing is the system's main structural safeguard, but it also means a single misplaced fraction propagates into both layers at once. This note records the audit of the manure nitrogen chain against the 2019 Refinement, the two implementation errors it found, and the regression tests that now pin the corrections. It is included because the corrected ledger described in the main text's methods is the outcome of that audit rather than an inherited assumption, and a reader should be able to see the evidence.

Defect 1: one volatilisation fraction used for two boundaries

The IPCC separates two volatilisation fractions that an earlier implementation of this work had conflated. $\mathrm{Frac_{GasMS}}$ (Table 10.22) is the nitrogen lost as NH_3 and NO_x *inside* the manure-management system — housing and storage, before manure reaches a field — and it is what reduces the manure available for application. $\mathrm{Frac_{GASM}}$ (Table 11.3) is the nitrogen lost *after* organic manure is applied to soil or deposited at grazing; under Tier 1 it drives indirect N_2O but does *not* reduce the organic nitrogen input before direct soil N_2O.

The earlier implementation used $\mathrm{Frac_{GASM}}$ at the pre-field boundary, where $\mathrm{Frac_{GasMS}}$ belongs, and then applied it a second time to the surviving nitrogen. One post-application fraction therefore stood in for a pre-field loss and was charged twice to the same nitrogen. The corrected chain applies the full Eq. 10.34a term at the management boundary with per-system Table 10.22 factors, and $\mathrm{Frac_{GASM}}$ exactly once, post-application. The effect on the national greenhouse-gas total is -0.62 Mt CO_2e, almost all of it in managed soils, because the organic nitrogen reaching cropland (F_{ON}) falls from 2.721 to 2.335 Mt N once the pre-field loss is charged at the correct boundary.

Defect 2: an ammonia share applied to a factor already in NH_3-N

The EMEP/EEA synthetic-fertiliser product factors are published as *ammonia nitrogen*. An earlier version multiplied the product-weighted value by a further NH_3 share of the volatilised pool, which understates ammonia and simultaneously leaves the IPCC indirect-N_2O pool short of its NO_x term, so the split stops conserving nitrogen. The three quantities are now defined and computed separately, as set out in the main text's air-pollutant methods: $f_{NH_3,SF} = 0.1608$ drives the ammonia estimate directly, $f_{NO,SF} = 0.007$ is a separate soil process, and only the IPCC indirect pathway uses their sum, $\mathrm{Frac_{GASF}} = 0.1678$, because Eq. 11.9 is defined on the combined volatilised pool. The net effect on the national total is $+0.13$ Mt CO_2e.

Diagnostic configurations

Because the two chains disagreed in aggregate before they disagreed visibly in any single stage, the audit compared them stage by stage on a common NH_3-N basis rather than on totals alone, and the resulting stage table is reported in the main text. That comparison showed the

similar national totals concealing substantial differences between individual stages. A totals-only comparison therefore did not establish agreement in nitrogen pathways.

Regression tests

The corrections are pinned rather than described. The test suite fixes the audited national total, the per-stage nitrogen balance on both bases, the identity $\mathrm{Frac}_{\mathrm{LossMS}} = \mathrm{Frac}_{\mathrm{GasMS}} + \mathrm{Frac}_{\mathrm{leachMS}} + \mathrm{Frac}_{\mathrm{N_2MS}} + EF_3$ per species and system, and the requirement that the manure-management scenarios in the sensitivity and structural-uncertainty tables be produced by the same full ledger rerun — a cross-module check added after re-slicing a single category produced two different national totals for one scenario in two tables of this paper. Every table in this manuscript is additionally regenerated from the engine and compared against the committed source, so a number cannot drift from the code that produced it without a test failing.

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